

ADKMNPT

COS214 PROJECT

PRESENTATION

2026

THE TEAM



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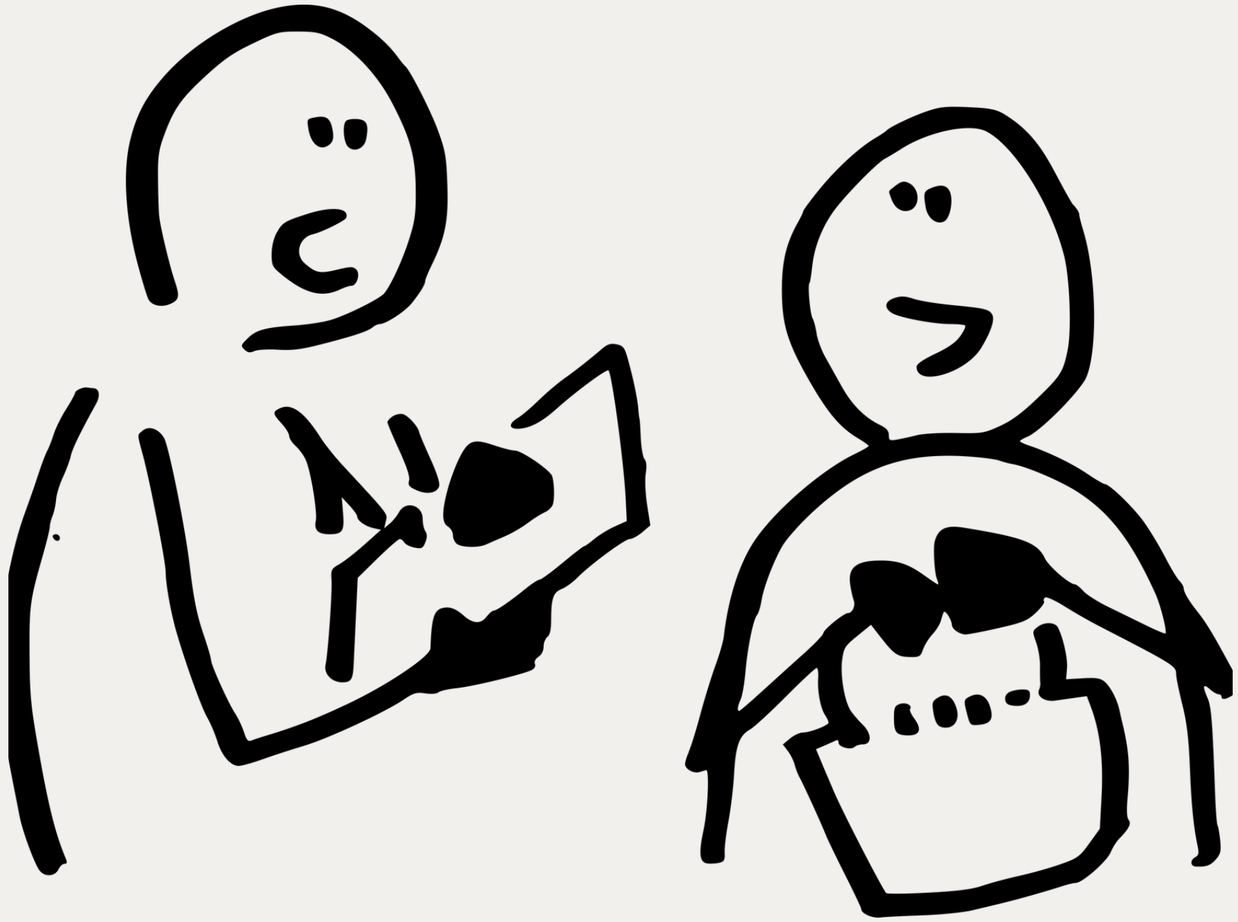
NTANDO DLAMINI



KATLEGO SENTSHO



AGREEMENT MOKGWATSANE



PROBLEM STATEMENT & SOLUTION

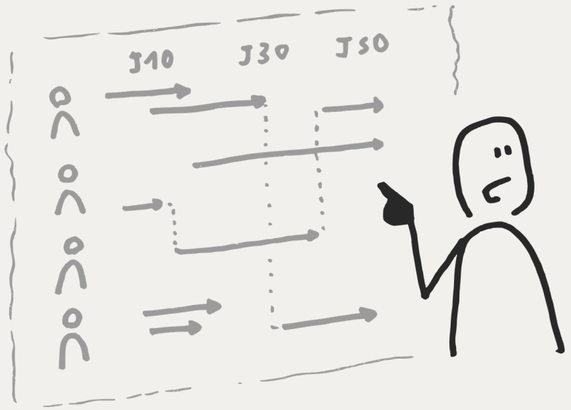
ORGANISATIONS NEED A STRUCTURED WAY TO COORDINATE TASKS, ENFORCE WORKFLOW RULES AND TRACK WORK WITHOUT RELYING ON DISCONNECTED MANUAL PROCESSES.

OUR SOLUTION IS TIVIDY

TIVIDY ?

- TIVIDY IS A GENERIC WORKFLOW MANAGEMENT SYSTEM
- IT MANAGES STRUCTURED BUSINESS PROCESSES INVOLVING TASKS, PARTICIPANTS, APPROVALS, DECISIONS AND DEPENDENCIES.
- IT ENABLES DIFFERENT INDUSTRIES TO EXECUTE WORKFLOWS USING THE SAME UNDERLYING ENGINE.
- IT TRACKS WORKFLOW PROGRESS, STATE CHANGES AND EXECUTION HISTORY.





OBJECTIVES

01 Develop a reusable, industry-independent workflow engine

02 Support workflow definitions and independent running instances

03 Manage work-item lifecycles, assignments and dependencies.

04 Demonstrate two different industries through one working C++ application

TIVIDY SYSTEM ARCHITECTURE

User -Interface (TUI)

User commands . Workflow status. Execution results

TividyWorkflowFacade

Single entry point to the workflow system

Generic TIVIDY Engine

WorkflowEngine + WorkflowDirector

Workflow Definitions

Builder • Composite

Execution & Control

State • Command • Mediator

Events & Monitoring

Observer • Singleton

Workflow Instances

InstanceData • Work Items

Routing & Approval

Strategy • Approval Chain

Recovery & Services

Memento • Adapter • Decorator

Scenario Plug-in Layer

Industry 1

Definition, rules, roles, data and adapters

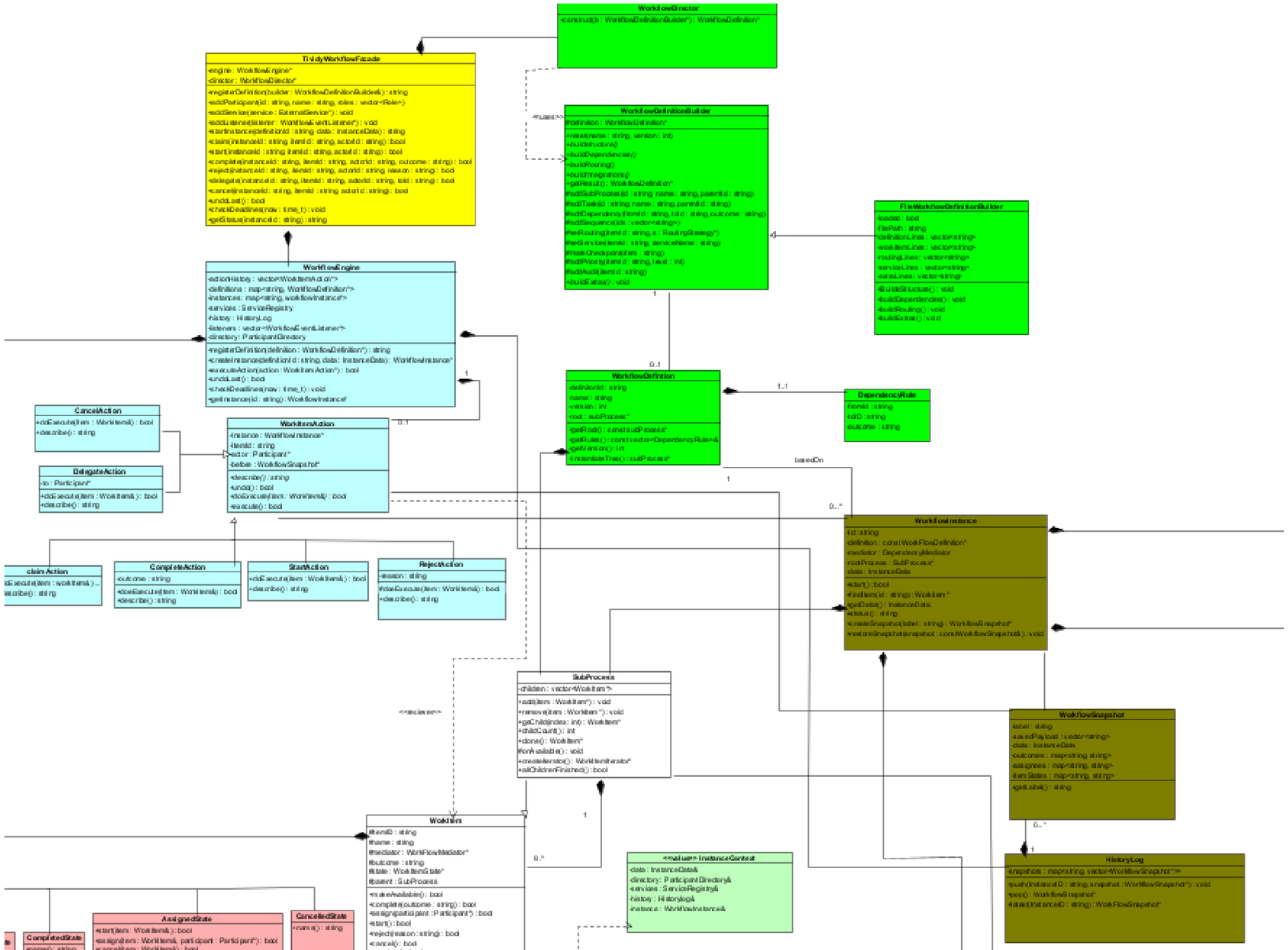
Industry 2

Definition, rules, roles,

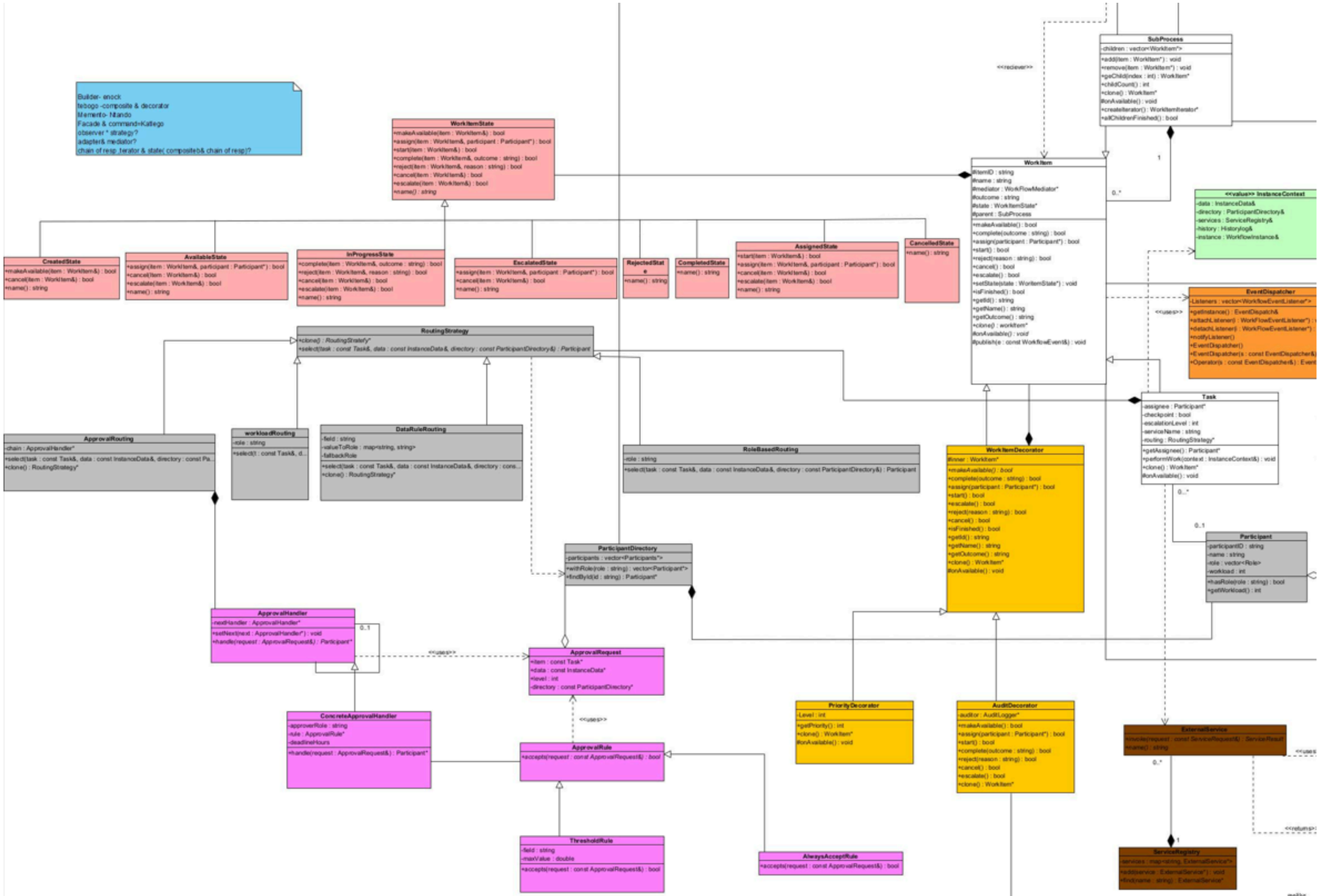
TIVIDY UML CLASS DIAGRAM



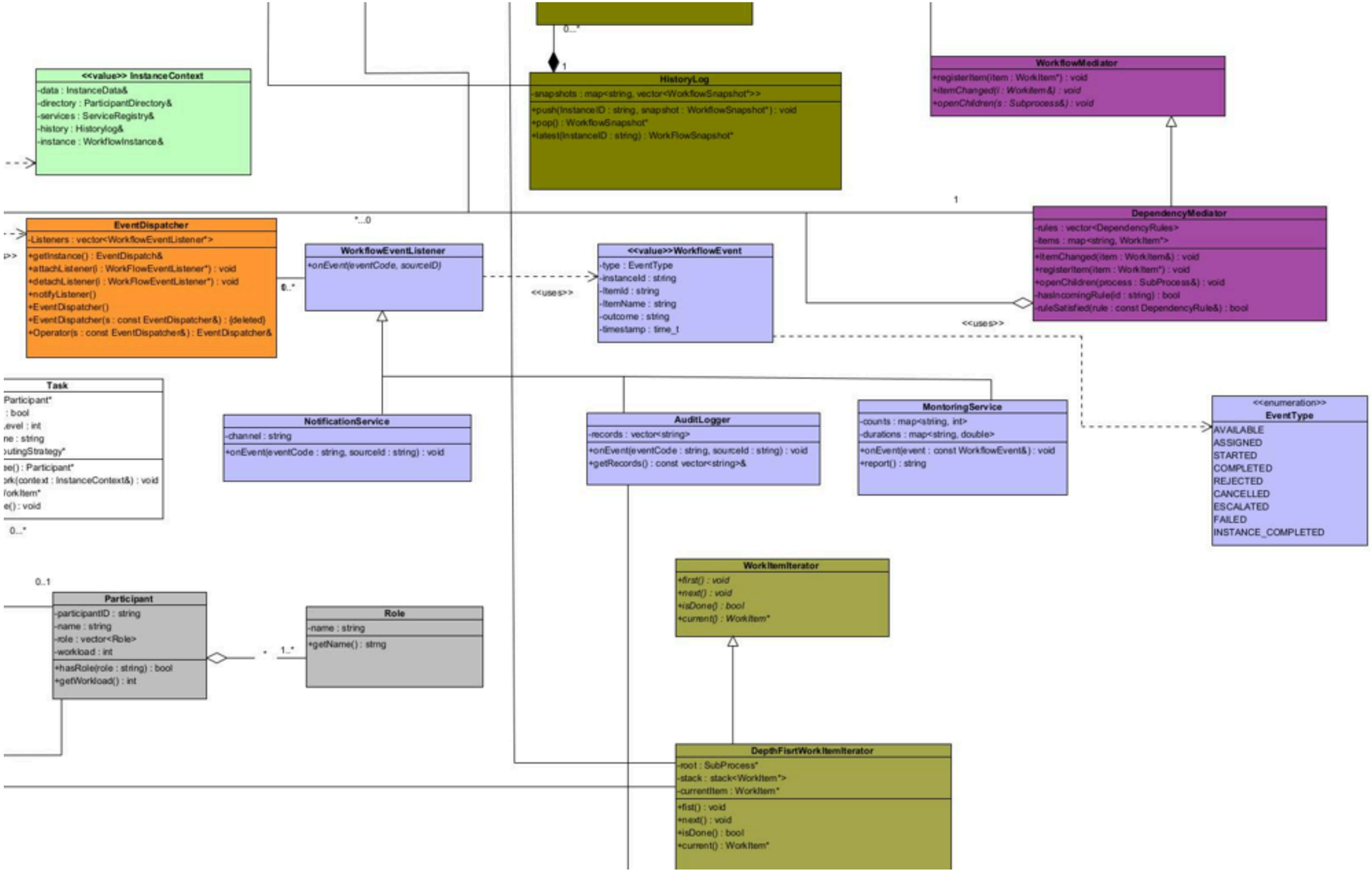
Facade, Command , Builder & Memento



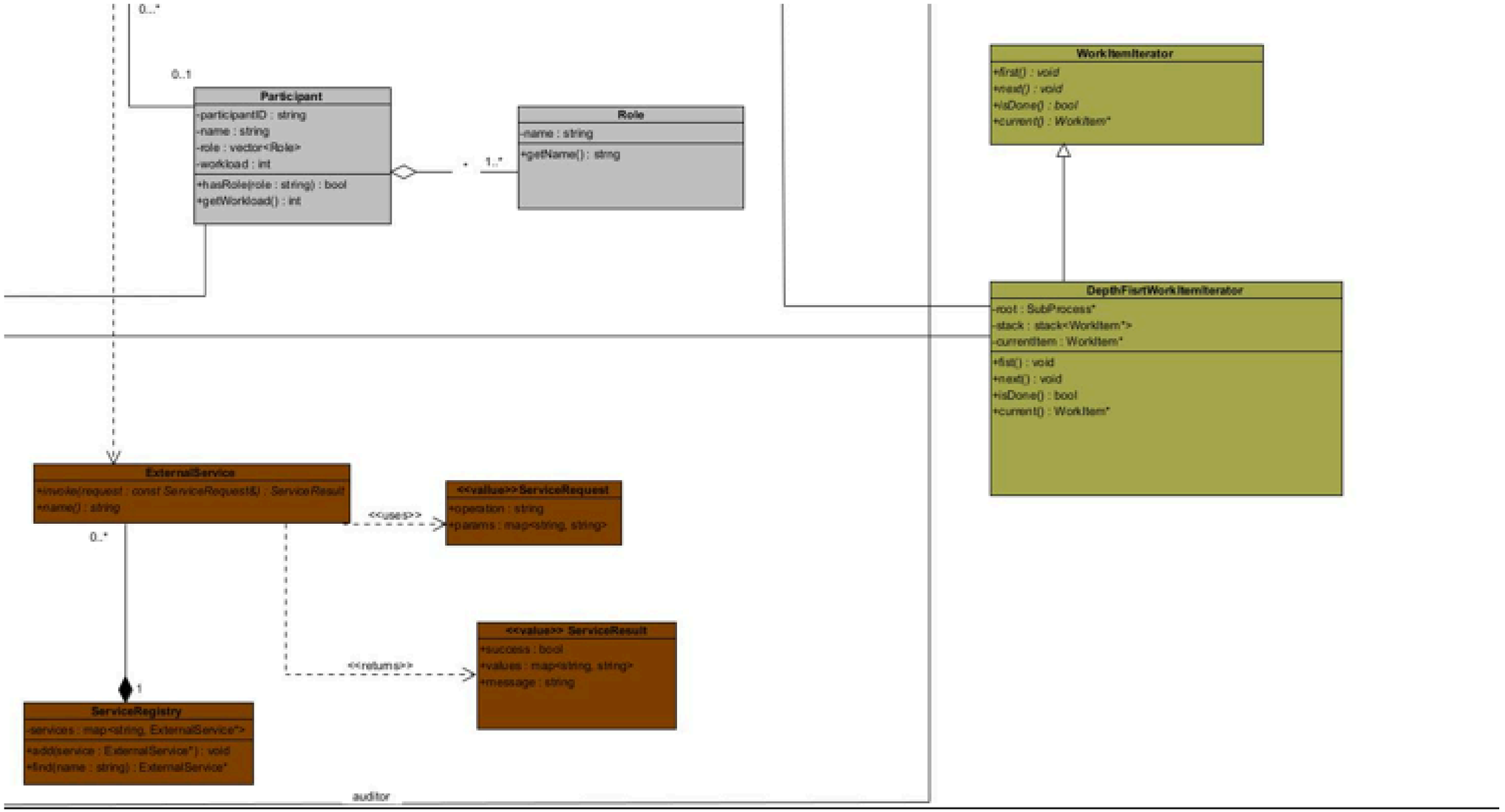
State, Composite,Decorator,Strategy& Chain of Responsibility



Observer,Singleton & Mediator

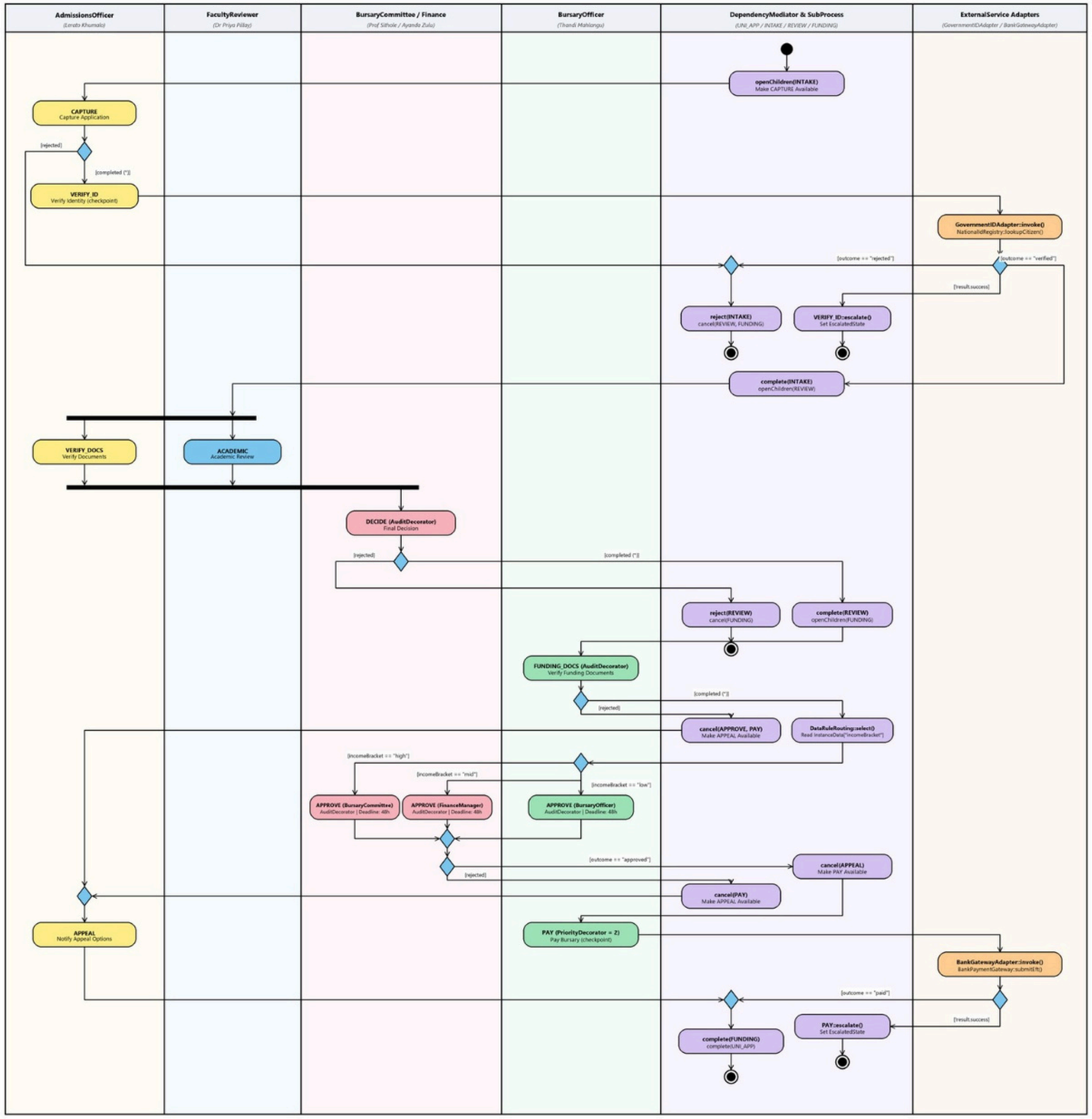


Iterator & Composite

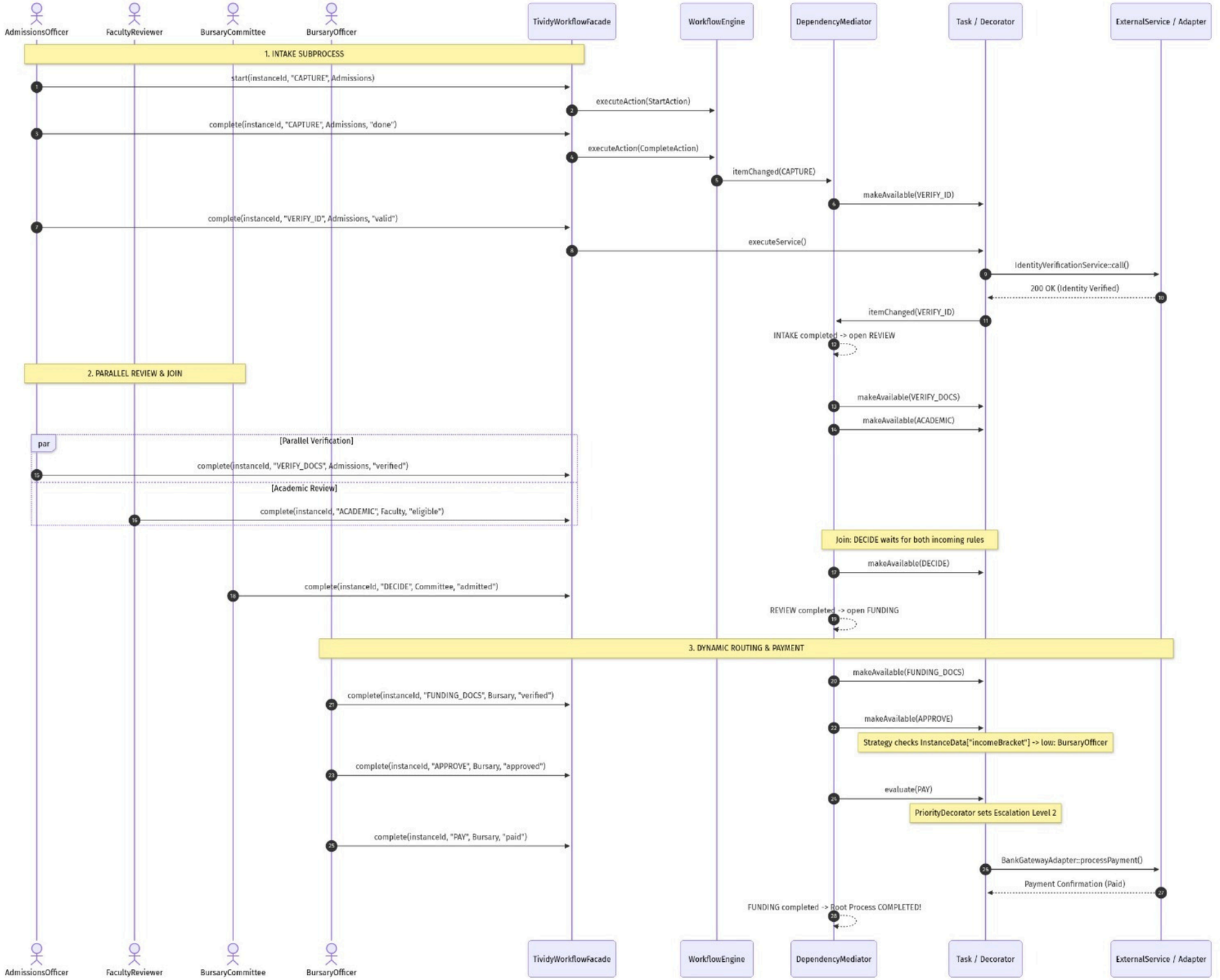


SCENARIO 1

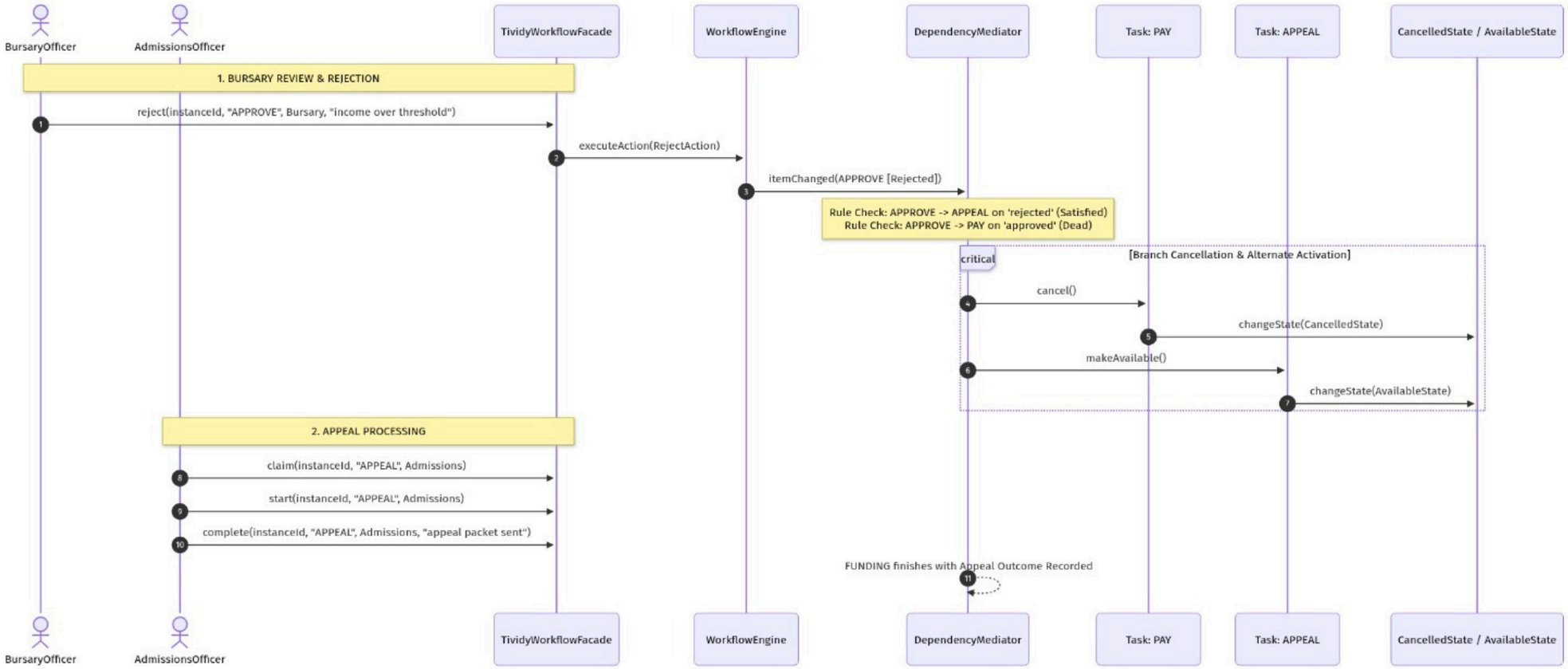
Scenario 1 Addmissions & bursary



Scenario 1 Addmissions & bursary



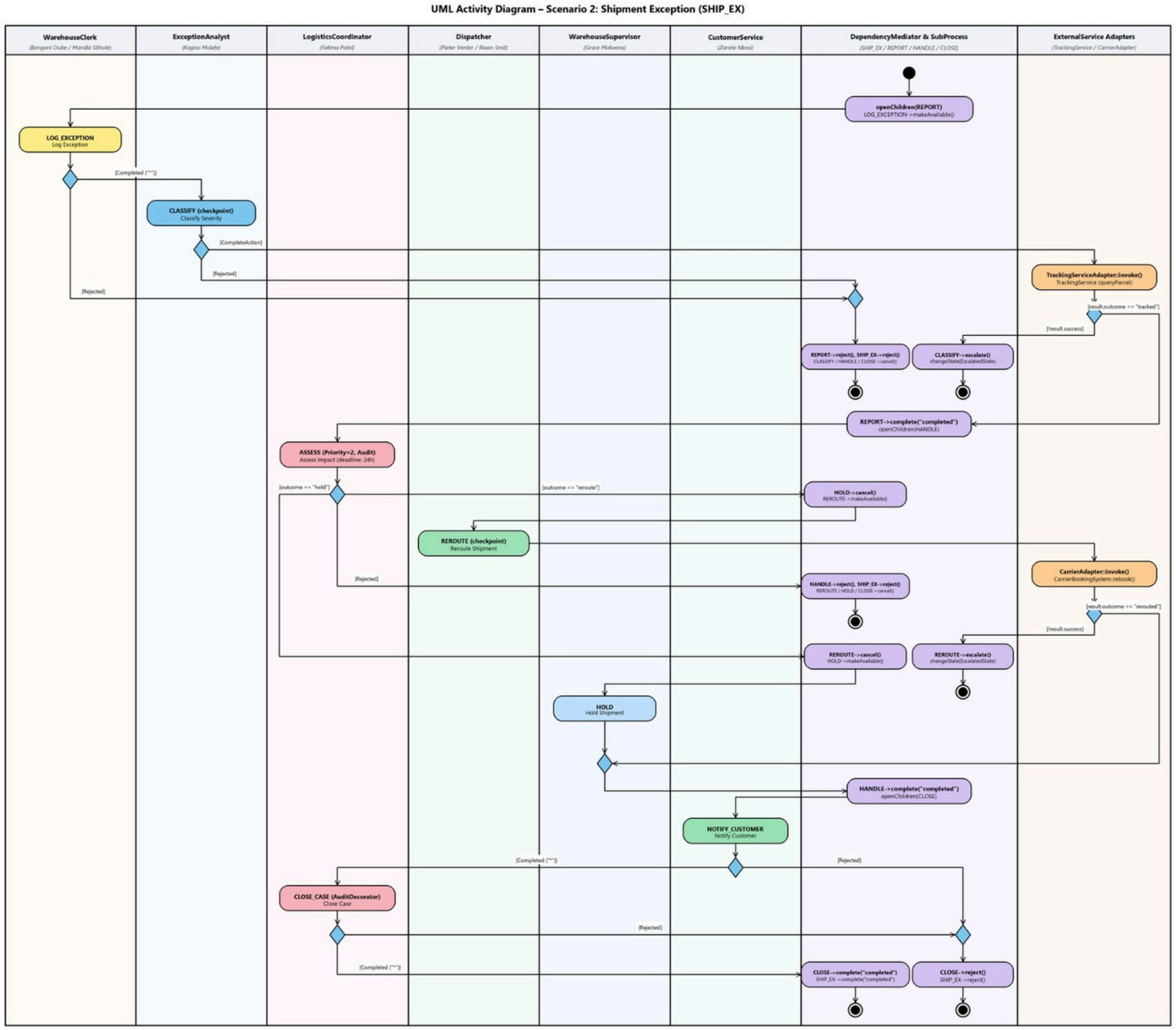
Failed Scenario 1



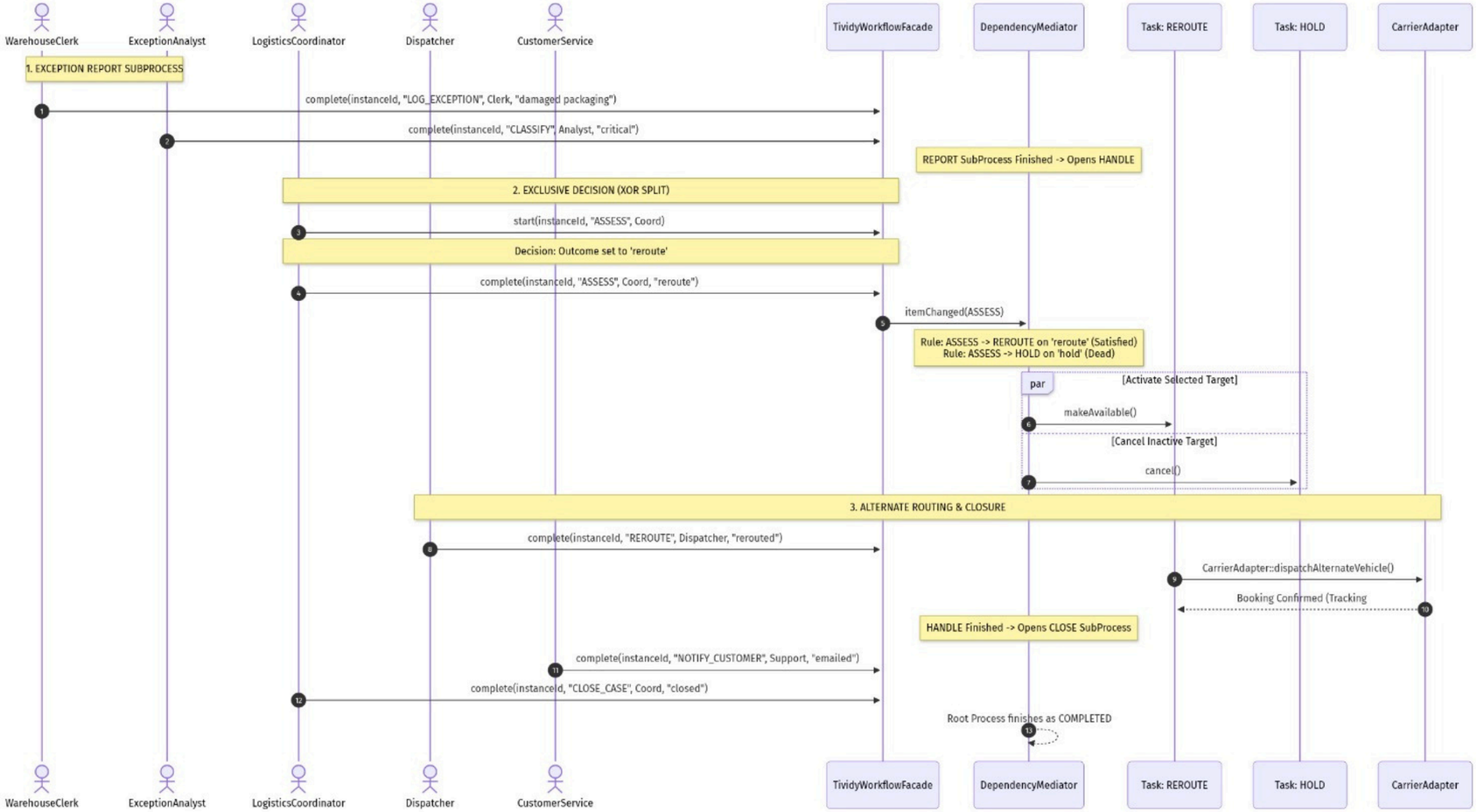
SCENARIO 2

Scenario 2

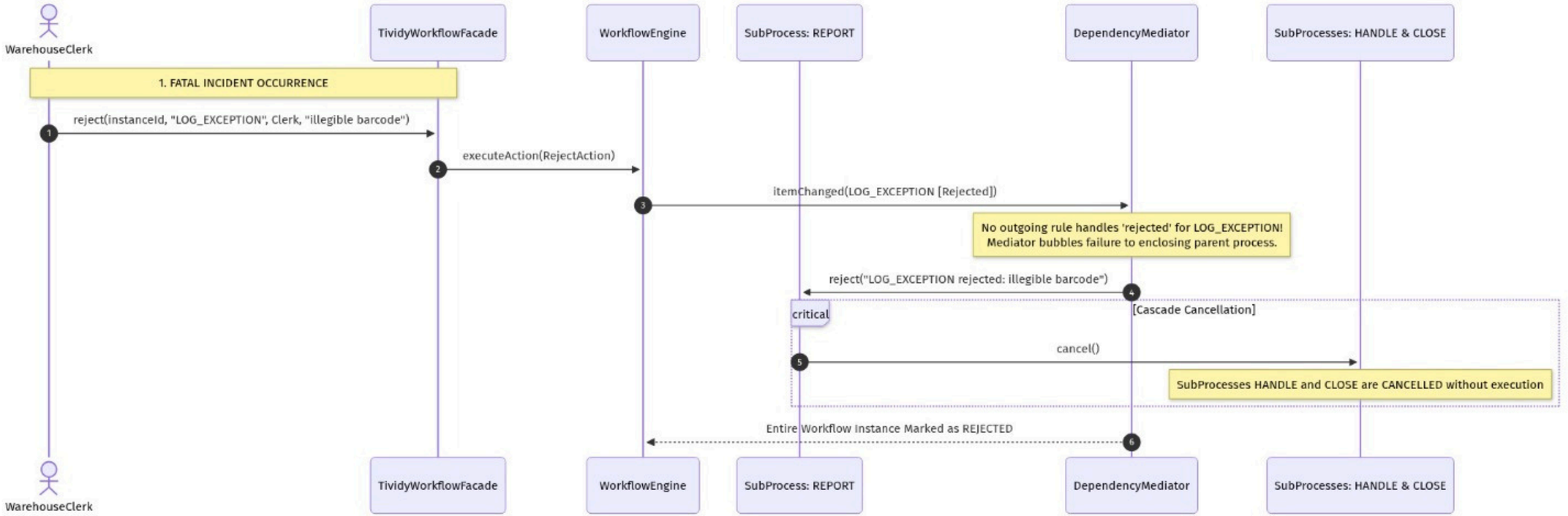
Logistics



Scenario 2 Logistics

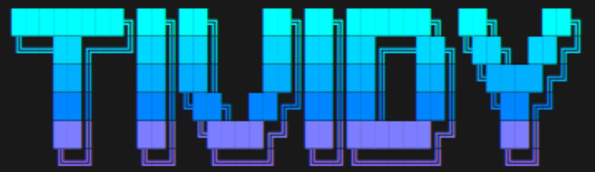


Failed Scenario 2



TUI

```
(base) ➜  COS214_P6 git:(dev) make run
./tividy
```



One workflow engine. **3 ready to run.**

Where are we going?

1	**Education**	University and Bursary Application
2	**Logistics**	Shipment Exception
3	**Something else**	SOMETHING ELSE
0	**Exit**	

❯ **Choose**: 1

TIVIDY › Education › Who are you?

Education participants

1	**Lerato Khumalo**	AdmissionsOfficer
2	**Dr Priya Pillay**	FacultyReviewer
3	**Prof Nomsa Sithole**	BursaryCommittee
4	**Thandi Mahlangu**	BursaryOfficer
5	**Ayanda Zulu**	FinanceManager
6	**Sipho Ndlovu**	WorkflowAdmin
7	**Naledi Dube**	AdmissionsOfficer
8	**Dr Johan Botha**	FacultyReviewer
9	**Rethabile Moeng**	BursaryOfficer
10	**Dr Sizwe Radebe**	EngineeringReviewer
0	**Back**	

❯ **Act as**: 1

TIVIDY › Education › University and Bursary Application

What just happened

✓ Now acting as Lerato Khumalo

```
┌ What just happened ──────────────────────────────────────────────────────────┐
✓ Now acting as Lerato Khumalo
└──────────────────────────────────────────────────────────────────────────────┘

┌ Main menu ───────────────────────────────────────────────────────────────────┐
Acting as Lerato Khumalo  AdmissionsOfficer

 1  Explain this workflow       what the process does and who is involved
 2  Start a workflow instance   enter the details and run it
 3  Open an instance            0 started, 0 running
 4  My work                     what you can act on right now
 5  Change participant          choose who you are acting as
 6  Undo the last action        roll back from a snapshot
 7  Full audit trail            every event, all workflows
 8  Monitoring metrics          live counts from the observers
 9  Check deadlines             jump ahead and see what escalates
 0  Back to the workflow list
└──────────────────────────────────────────────────────────────────────────────┘

❯ Choose: 2
  TIVIDY  ›  Education  ›  Start a workflow instance

┌ New instance, started by Lerato Khumalo ─────────────────────────────────────┐
The engine gives it an instance id. Press Enter on an empty line to go back.
└──────────────────────────────────────────────────────────────────────────────┘

❯ Applicant surname: Naidoo
❯ Applicant ID number, 13 digits: 0403120123081
❯ Degree applied for (Engineering goes to the engineering reviewer): BEng Computer Engineering
  TIVIDY  ›  instance-1  ›  Education
```

```
New instance, started by Lerato Khumalo
The engine gives it an instance id. Press Enter on an empty line to go back.

❯ Applicant surname: Naidoo
❯ Applicant ID number, 13 digits: 0403120123081
❯ Degree applied for (Engineering goes to the engineering reviewer): BEng Computer Engineering
  TIVIDY  ›  instance-1  ›  Education

What just happened
✓ Started instance-1 by Lerato Khumalo
AVAILABLE  University and Bursary Applic...
STARTED    University and Bursary Applic...
AVAILABLE  Application Intake
STARTED    Application Intake
AVAILABLE  Capture Application
ASSIGNED   Capture Application              -> Lerato Khumalo

University and Bursary Application
Started by Lerato Khumalo, Naidoo, 0403120123081, BEng Computer Engineering
RUNNING                                0/9 tasks finished
Acting as Lerato Khumalo  AdmissionsOfficer

  #   Work item                          State         Assignee          Outcome
  1   University and Bursary Application ▶ InProgress  -
  2   ├─ Application Intake              ▶ InProgress  -
▸ 3   │  ├─ Capture Application          ● Assigned    Lerato Khumalo
  4   │  └─ Verify Identity              ○ Created     -
  5   ├─ Application Review              ○ Created     -
  6   │  ├─ Verify Documents            ○ Created     -
  7   │  ├─ Academic Review             ○ Created     -
  8   │  └─ Final Decision              ○ Created     -
  9   └─ Bursary Funding                 ○ Created     -
  10     ├─ Verify Funding Documents     ○ Created     -
  11     ├─ Approve Bursary              ○ Created     -
  12     ├─ Pay Bursary ★2               ○ Created     -
  13     └─ Notify Appeal Options        ○ Created     -

▸ you can act on this now    ★ priority    purple stages contain tasks

  Number open a work item  h history  d data  c commands  p change participant  u undo  b back
❯ Choose:
```

Actions

Acting as **Lerato Khumalo** AdmissionsOfficer

1	Claim	the engine will refuse this
2	Start work	the engine will refuse this
3	**Complete**	✓ **allowed**
4	**Reject**	✓ **allowed**
5	Delegate to someone else	the engine will refuse this
6	**Cancel**	✓ **allowed**
0	**Back**	

❯ **Choose**: 3
❯ **Outcome [done]**: completed
TIVIDY › instance-1 › Education

What just happened

✓ Completed Capture Application (completed) by Lerato Khumalo
COMPLETED Capture Application completed
AVAILABLE Verify Identity
ASSIGNED Verify Identity -> Lerato Khumalo

University and Bursary Application

Started by Lerato Khumalo, Naidoo, 0403120123081, BEng Computer Engineering
RUNNING **1/9** tasks finished
Acting as **Lerato Khumalo** AdmissionsOfficer

#	Work item	State	Assignee	Outcome
1	**University and Bursary Application**	▶ **InProgress**	-	
2	├─ **Application Intake**	▶ **InProgress**	-	
3	│ ├─ Capture Application	✓ **Completed**	Lerato Khumalo	completed
▸ 4	│ └─ **Verify Identity**	● **Assigned**	Lerato Khumalo	
5	├─ **Application Review**	○ Created	-	
6	│ ├─ Verify Documents	○ Created	-	
7	│ ├─ Academic Review	○ Created	-	
8	│ └─ Final Decision	○ Created	-	
9	└─ **Bursary Funding**	○ Created	-	
10	├─ Verify Funding Documents	○ Created	-	
11	├─ Approve Bursary	○ Created	-	
12	├─ Pay Bursary ★2	○ Created	-	

University and Bursary Application

Started by Lerato Khumalo, Naidoo, 0403120123081, BEng Computer Engineering

RUNNING ████████ **3/9** tasks finished

Acting as **Lerato Khumalo** AdmissionsOfficer

#	Work item	State	Assignee	Outcome
1	University and Bursary Application	▶ InProgress	-	
2	├─ Application Intake	✓ Completed	-	completed
3	│ ├─ Capture Application	✓ Completed	Lerato Khumalo	completed
4	│ └─ Verify Identity	✓ Completed	Lerato Khumalo	verified
5	├─ Application Review	▶ InProgress	-	
6	│ ├─ Verify Documents	✓ Completed	Lerato Khumalo	completed
7	│ ├─ Academic Review	● Assigned	Dr Priya Pillay	
8	│ └─ Final Decision	○ Created	-	
9	└─ Bursary Funding	○ Created	-	
10	├─ Verify Funding Documents	○ Created	-	
11	├─ Approve Bursary	○ Created	-	
12	├─ Pay Bursary ★2	○ Created	-	
13	└─ Notify Appeal Options	○ Created	-	

▸ you can act on this now ★ priority **purple** stages contain tasks

Number open a work item **h** history **d** data **c** commands **p** change participant **u** undo **b** back

❯ **Choose**: d

TIVIDY › Data of instance-1

Instance data

degree	BEng Computer Engineering
idNumber	0403120123081
identityStatus	VERIFIED
identityVerified	true
instanceId	instance-1
startedBy	Lerato Khumalo
surname	Naidoo

External service replies

VERIFY_ID.message	Identity verified
VERIFY_ID.outcome	verified
VERIFY_ID.success	true

```
┌ What just happened ──────────────────────────────────────────────────────────┐
│ ✓ Completed Final Decision (approved) by Prof Nomsa Sithole                  │
│ COMPLETED  Final Decision                  approved                          │
│ AVAILABLE  Bursary Funding                                                   │
│ STARTED    Bursary Funding                                                   │
│ AVAILABLE  Verify Funding Documents                                          │
│ ASSIGNED   Verify Funding Documents        -> Thandi Mahlangu                │
│ COMPLETED  Application Review              completed                         │
└──────────────────────────────────────────────────────────────────────────────┘

┌ University and Bursary Application ──────────────────────────────────────────┐
│ Started by Lerato Khumalo, Naidoo, 0403120123081, BEng Computer Engineering  │
│ RUNNING  ████████████████████  5/9 tasks finished                            │
│ Acting as Prof Nomsa Sithole  BursaryCommittee                               │
│                                                                              │
│   #   Work item                          State         Assignee          Outcome
│   1   University and Bursary Application ▶ InProgress  -
│   2   ├─ Application Intake              ✓ Completed   -                 completed
│   3   │  ├─ Capture Application          ✓ Completed   Lerato Khumalo    completed
│   4   │  └─ Verify Identity              ✓ Completed   Lerato Khumalo    verified
│   5   ├─ Application Review               ✓ Completed   -                 completed
│   6   │  ├─ Verify Documents             ✓ Completed   Lerato Khumalo    completed
│   7   │  ├─ Academic Review              ✓ Completed   Dr Priya Pillay   sdfghfd
│   8   │  └─ Final Decision               ✓ Completed   Prof Nomsa Sithole approved
│   9   └─ Bursary Funding                  ▶ InProgress  -
│   10      ├─ Verify Funding Documents     ● Assigned    Thandi Mahlangu
│   11      ├─ Approve Bursary              ○ Created     -
│   12      ├─ Pay Bursary ★2               ○ Created     -
│   13      └─ Notify Appeal Options        ○ Created     -
│                                                                              │
│ ▸ you can act on this now    ★ priority    purple stages contain tasks       │
└──────────────────────────────────────────────────────────────────────────────┘
  Number open a work item  h history  d data  c commands  p change participant  u undo  b back
❯ Choose:
```

University and Bursary Application

Started by Lerato Khumalo, Naidoo, 0403120123081, CS
RUNNING ████████████████ 6/9 tasks finished
Acting as **Thandi Mahlangu** BursaryOfficer

#	Work item	State	Assignee	Outcome
1	University and Bursary Application	▶ InProgress	-	
2	├─ Application Intake	✓ Completed	-	completed
3	│ ├─ Capture Application	✓ Completed	Lerato Khumalo	done
4	│ └─ Verify Identity	✓ Completed	Lerato Khumalo	verified
5	├─ Application Review	✓ Completed	-	completed
6	│ ├─ Verify Documents	✓ Completed	Lerato Khumalo	done
7	│ ├─ Academic Review	✓ Completed	Dr Priya Pillay	done
8	│ └─ Final Decision	✓ Completed	Prof Nomsa Sithole	approved
9	└─ Bursary Funding	▶ InProgress	-	
10	├─ Verify Funding Documents	✓ Completed	Thandi Mahlangu	approved
▸ 11	├─ **Approve Bursary**	▶ InProgress	Thandi Mahlangu	
12	├─ Pay Bursary ★2	○ Created	-	
13	└─ Notify Appeal Options	○ Created	-	

▸ you can act on this now ★ priority **purple** stages contain tasks

Number open a work item **h** history **d** data **c** commands **p** change participant **u** undo **b** back
❯ **Choose:** 11
TIVIDY › instance-2 › Approve Bursary

Approve Bursary (APPROVE)

State ▶ InProgress
Assignee Thandi Mahlangu
Deadline 48 hours
Approval ▸ **BursaryOfficer (amount <= 50000)**
FinanceManager (amount <= 150000)
BursaryCommittee (any request)

Actions

Acting as **Thandi Mahlangu** BursaryOfficer

1	Claim	the engine will refuse this
2	Start work	the engine will refuse this
3	**Complete**	✓ **allowed**
4	**Reject**	✓ **allowed**
5	Delegate to someone else	the engine will refuse this
6	**Cancel**	✓ **allowed**
0	**Back**	

❯ **Choose:** 3
1 **approved**
2 **not approved**
❯ **Outcome:** 1
TIVIDY › instance-2 › Education

What just happened

✓ Completed Approve Bursary (approved) by Thandi Mahlangu
COMPLETED Approve Bursary approved
AVAILABLE Pay Bursary

```
State       ▶ InProgress
Assignee    Thandi Mahlangu
Priority    ★ 2
Role        BursaryOfficer
Service     PaymentGateway  (called when you complete this)
```

```
Actions
Acting as Thandi Mahlangu  BursaryOfficer

 1  Claim                      the engine will refuse this
 2  Start work                 the engine will refuse this
 3  Complete                   ✓ allowed
 4  Reject                     ✓ allowed
 5  Delegate to someone else   the engine will refuse this
 6  Cancel                     ✓ allowed
 0  Back
```

```
❯ Choose: 3
❯ Outcome [done]: payed
  TIVIDY  ›  instance-2  ›  Education
```

```
What just happened
✓ Completed Pay Bursary (paid) by Thandi Mahlangu
⇄ PaymentGateway: Payment EFT100001 accepted
COMPLETED  Pay Bursary                    paid
COMPLETED  Bursary Funding                completed
COMPLETED  University and Bursary Applic... completed
```

```
University and Bursary Application
Started by Lerato Khumalo, Naidoo, 0403120123081, CS
COMPLETED  ████████████████████ 9/9 tasks finished
Acting as Thandi Mahlangu  BursaryOfficer

  #   Work item                          State         Assignee            Outcome
  1   University and Bursary Application ✓ Completed   -                   completed
  2   ├─ Application Intake              ✓ Completed   -                   completed
  3   │  ├─ Capture Application          ✓ Completed   Lerato Khumalo      done
  4   │  └─ Verify Identity              ✓ Completed   Lerato Khumalo      verified
  5   ├─ Application Review              ✓ Completed   -                   completed
  6   │  ├─ Verify Documents            ✓ Completed   Lerato Khumalo      done
  7   │  ├─ Academic Review             ✓ Completed   Dr Priya Pillay     done
  8   │  └─ Final Decision              ✓ Completed   Prof Nomsa Sithole  approved
  9   └─ Bursary Funding                 ✓ Completed   -                   completed
 10      ├─ Verify Funding Documents     ✓ Completed   Thandi Mahlangu     approved
 11      ├─ Approve Bursary              ✓ Completed   Thandi Mahlangu     approved
 12      ├─ Pay Bursary ★2               ✓ Completed   Thandi Mahlangu     paid
 13      └─ Notify Appeal Options        - Cancelled   -

▸ you can act on this now    ★ priority    purple stages contain tasks
```

```
  Number open a work item  h history  d data  c commands  p change participant  u undo  b back
❯ Choose:
```

CONCLUSION

GITHUB & BRANCHING

MEMORY MANAGEMENT

DOCKER

DOCTEST

DOXYGEN

COLLABORATION

THANK YOU!



